

SUMMER TRAINING PROJECT REPORT



पूर्णतः सहकारी स्वामित्व

Wholly owned by Cooperatives

INDIAN FARMERS FERTILIZER COOPERATIVE LIMITED

PARADEEP UNIT



This plant training report has been submitted as partial fulfillment of 4-year B. Tech in Chemical Engineering.

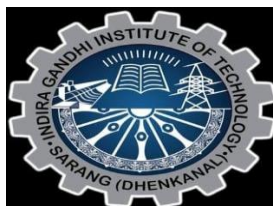
Submitted by

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Bachelor in Technology, 6th semester

Duration- 1st July 2024 to 30th July 2024

**INDIRA GANDHI INSTITUTE OF TECHNOLOGY, SARANG,
DHENKANAL, ODISHA**



CERTIFICATE

This is to certify that the industrial training report has been submitted by **PUNAM MOHAPATRA**, Branch **Chemical Engineering**, Institute **IGIT SARANG** under my guidance and supervision has completed the industrial training at IFFCO, PARADEEP unit

Submitted to
Mr K.P NANDA sir
Training Officer

Submitted by
Punam Mohapatra
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ACKNOWLEDGEMENT

I express my sincere thanks and gratitude to IFFCO Paradeep Unit for giving me the opportunity for completing our in-plant summer training at Phosphoric Acid Plant. I extend my appreciation to all the employees of IFFCO Paradeep Unit who generously shared their time, knowledge, and perspectives with me during the internship. My vocabulary falls short of words to express my sincere gratitude to Mr. Kamakshi Prasad Nanda (Chief Manager, Training) under whose guidance I had the opportunity to carry out the present work. I would like to express my gratitude towards all officers, managers, management trainees & attendants of IFFCO. For their excellent support, hospitality & proper guidance in completing my internship research and formulating report.

I am also thankful to my academic institution 'Indira Gandhi Institute of Technology, Sarang' for providing me with the opportunity to undertake this internship, which has been pivotal in my professional development. The theoretical foundation provided by my coursework was complemented by practical exposure at IFFCO, thereby bridging the gap between classroom learning and real-world applications.

Last but not the least, our librarian Mr. Barik Sir who provided us every material we needed. Other sources include manuals from the library and internet sites to know few details.

Lastly, I would like to acknowledge my family and friends for their unwavering support and encouragement throughout this endeavour.

This report aims to encapsulate the experience and reflections gained during my internship, serving as a record of my growth and learning during this enriching period.

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ABOUT IFFCO

INDIAN FARMERS' FERTILIZER COOPERATIVE LIMITED is the largest fertilizer cooperative in the world and is wholly owned Cooperatives. The company was established on 3rd November 1967. IFFCO got deemed recognition under the provisions of the Multistate Cooperative Societies Act 1984 & 2002 later. Recently in the year 2017, IFFCO completed its Golden Jubilee Year.

Currently, Dr. Udai Shanker Awasthi is the Managing Director and Chief Executive Officer of IFFCO. He took charge in 1993.

IFFCO has 35,000 member cooperatives. It has been registered as a multi-state cooperative society. Society is primarily engaged in the production and distribution of fertilizers.

❖ IFFCO has 5 units all over India namely: -

- Kalol and Kandla unit in Gujarat.
- Aolla and Phulpur unit in Uttar Pradesh.
- Paradeep unit in Orrisa.

❖ In addition to this IFFCO has units in foreign namely: -

- Indo-Egyptian Fertilizer Company, in Egypt.
- Jordan-India Fertilizer Company, in Jordan.
- Iran-India Fertilizer Company, in Oman.

❖ In addition to the fertilizer plants IFFCO has different ventures namely: -

- IFFCO Chhattisgarh Power Limited.
- IFFCO Kisan Sanchar Limited.
- IFFCO-Tokio General Insurance. (IFFCO's own insurance company)

MISSION

- IFFCO's mission is "to enable Indian farmers to prosper through timely supply of reliable, high quality agricultural inputs and services in an environmentally sustainable manner and to undertake other activities to improve their welfare".
- To provide to farmers high-quality fertilisers in the right time and in adequate quantities to increase crop productivity.
- Commitment to health, safety, environment and forestry development to enrich the quality of community life.
- To institutionalise core values and create a culture of team building, empowerment and innovation which would help in the incremental growth of employees and enable achievement of strategic objectives.
- Foster a culture of trust, openness and mutual concern to make working, a stimulating and challenging experience for stakeholders.
- To acquire, assimilate and adopt reliable, efficient and cost-effective technologies.
- A true cooperative society committed for fostering cooperative movement in the country. Emerging as a dynamic organisation, focussing on strategic strengths, seizing opportunities for generating and building upon past success, enhancing earnings to maximise the shareholders' value.
- To make plants energy efficient and continually review various schemes to conserve energy.
- Sourcing raw materials for the production of phosphatic fertilisers at economical cost by entering into Joint Ventures outside India.
- Building a value driven organisation with an improved and responsive customer focus. A true commitment to transparency, accountability and integrity in principle and practice.
- Commitment to social responsibilities for a strong social fabric.
- To ensure growth in core and non-core sectors.

VISION

- To stand as a Global Leader in Fertilisers production
- Implement strategies for sustainable development by reducing Energy Consumption and better Resource Management
- Maximizing the synergies of Core Business through Forward/Backward Integrations
- Enhancing presence in International markets through Strategic Joint Ventures and Synergistic Acquisitions
- Diversification in other Sectors for financial sustainability
- Promoting Integrated Nutrient Management and optimum Fertiliser use
- To help the Cooperative Societies become economically strong, professionally managed and to equip the Farming Community with advanced agricultural practices for improved Productivity, to ensure an Empowered Rural India
- Achieve Fertilisers Marketing target of 15 million tonnes per annum

ABOUT THE IFFCO - PARADEEP UNIT

- The Paradeep unit was established in the year 1998 by "OSWALD CHEMICALS & FERTILIZERS LIMITED".
- The unit was designed by LURGES COMPANY of the USA.
- Then OCFL started its production in the year 2000. They were able to run the plant till 2005.
- Due to different environmental and pollution problems, OCFL was not able to run the plant and handed over the plant to IFFCO in October 2005.

- The unit is spread over an area of 2077 acres and is an integral fertilizer complex having the world's single largest phosphoric acid plant.

<u>Name Unit</u>	<u>No. of Units</u>	<u>Average Capacity</u>
○ Phosphoric Acid Plant	01	2650 MTPD
○ Sulphuric Acid Plan	02	3500 MTPD for each unit
○ DAP Plant	3 Trains	2000 MTPD for each Train
○ Captive Power Plant	02	55 MW for each unit

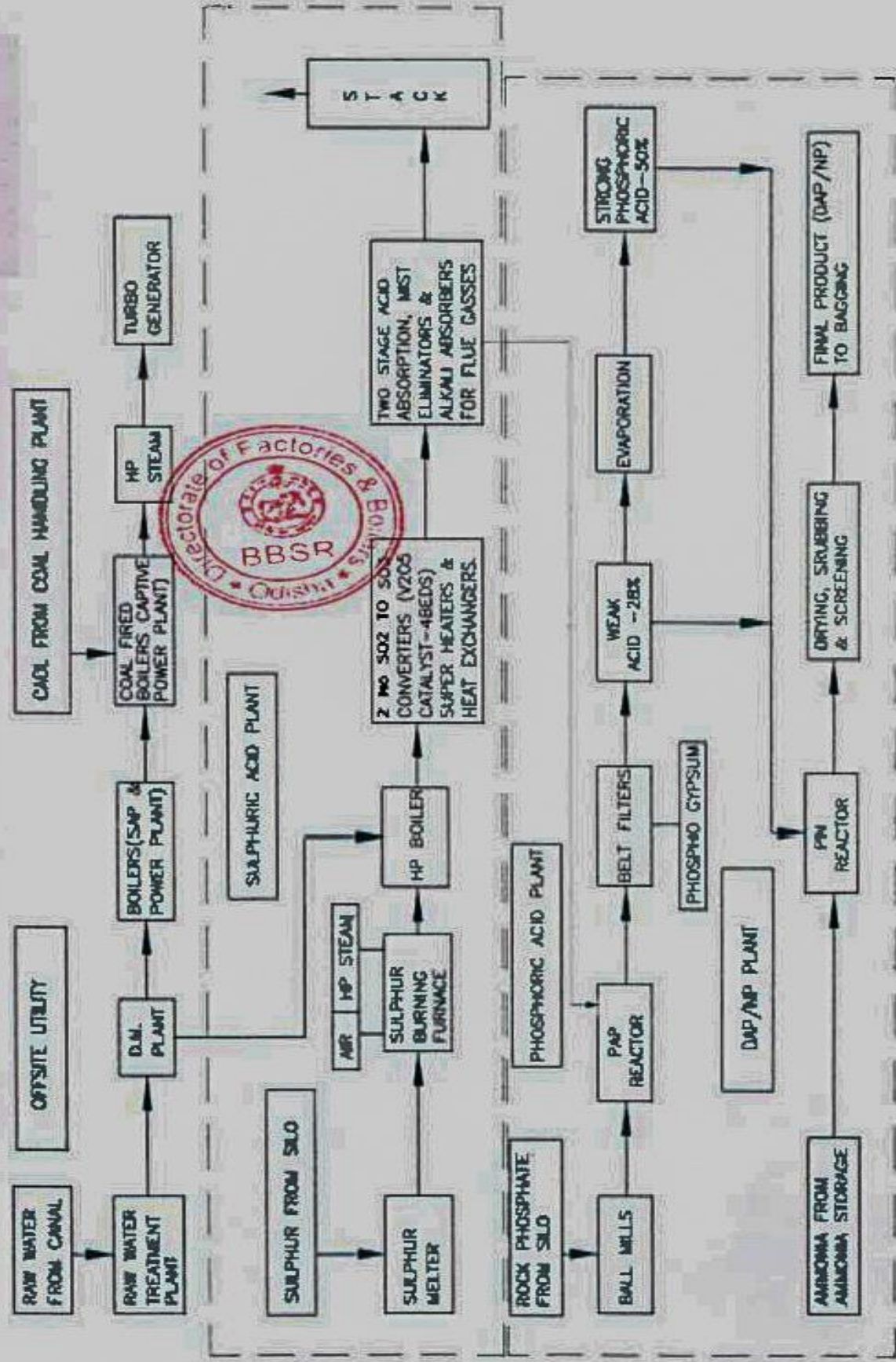
- The Paradeep Unit also has a DM Plant, Compression House, Effluent Treatment Plant, Sewerage Treatment Plant, and a Utility Offsite.
- Under the U&O the unit has a provision for storing Ammonia. For the storage of ammonia, the unit has 3 tanks-2 tanks of 20,000 MT each and 1 tank of 10,000 MT.
- The unit has its jetty in the Paradeep Port. All the raw materials such as phosphates, sulfurs, ammonia, and potash come from outside India and are unloaded in the jetty and are carried by a conveyor system to the plant raw material silos. For transportation of ammonia, a pipeline arrangement is made from the port to the plant's U&O.
- The unit has a strong marketing unit all over India in almost all states.
- The unit has a "Bagging Plant" where there are 14 chutes for filling the field bags.
- For transportation, the unit has 150 trucks. 3kms away from the plant, the bagged products are transported to a place named Rangaiah Garh, where there is a railway cargo terminal from where the finished goods are transported all over India.

<u>Raw Material</u>	<u>Use</u>	<u>Place of Import</u>
Rock Phosphate.	Produce HPO	Jordan, Egypt, Peru
Sulfur	Produce H ₂ SO ₄	Saudi Arabia, Kuwait, Iraq, Iran
Ammonia	Produce DAP	From Middle East countries
Muriate of Potash (KCl)	Produce NPK-1.NPK-11.	Canada and Peru



PROCESS FLOW DIAGRAM OF CPP

PROCESS FLOW DIAGRAM (IFFCO PARADEEP PLANT)



FIRE AND SAFETY AT IFFCO - PARADEEP UNIT

The chemical industry is fast-growing and new technological development is taking place day by day. Modern chemical plants are equipped with sophisticated controlling devices: however, one has to face risk, during the operation or maintenance of the plant/equipment. The manufacture of the fertilizer in the Paradeep unit is based on the shipment of Rock Phosphate, Sulphur, and Ammonia through cross-country conveyors and pipelines from the port jetty. To ensure safe working, the supervisors and the workers must be trained in safe working procedures, safety rules, and knowledge of the statutory obligations. Safety is the first step that is being followed in the IFFCO Paradeep unit.

In a fertilizer industry extremely dangerous/toxic, highly inflammable, and highly volatile chemicals/materials are handled. Therefore, the emphasis on fire prevention, training, and rigid enforcement of fire prevention regulation is required. It is a fact that most fire occurs due to human failure and due to non-compliance with established procedures and recognized safe practices.

General Safety Rules:

- Employees must report all accidents immediately to their supervisor.
- Any defective or unsafe equipment must be reported to a supervisor so that it may be repaired promptly.
- Conform to all safety rules, signs, and regulations. If the rules are not understood then the supervisor shall explain them to them.
- A safety belt with a life line properly attached, anchored, and adjusted must be worn when working at locations without guardrails, or substantial barriers, i.e., the location is more than 2 meters above the lower floor or ground level.
- Access to the fire extinguisher or other firefighter equipment must not be blocked.
- Machine guard should be in position before start-up or during normal operation of any equipment.
- Electrical repairing must only be made by authorized personnel.
- Makeshift scaffolds, ladders, ropes, blocks, cables, or defective slings must not be used.

- Compressed air must not be used to blow off any part of the body or any clothing while it is being worn, except at authorized locations.
- Use pull chords in case of emergency with running conveyor belts.
- Wear helmets and shoes on factory premises.

Safety Measures followed inside the Plant:

- Speed control on roads. (20 km/hr)
- Specific measures adopted for work at height, e.g., hydraulic elevated scissor lift.
- "SURAKSHA SABHA" is conducted for all contract workers.
- Height Phobia Test is conducted for every safety personnel



SECTION:- 1

PHOSPHORIC ACID PLANT

1.1 Introduction:-

The production of Phosphoric acid plays an important role in the manufacturing of Phosphatic fertilizers. Usually, ally, the plant requires supplements like Nitrogen, Phosphorus, and potassium for its growth. Nitrogenous fertilizer like Urea supplements only nitrogen in the soil. But as such soil also requires phosphorous element for its growth, phosphorous element supplement can be given to soil by the application of Phosphatic Fertilizers (N-P-K-S) N stands for Nitrogen, P stands for P_2O_5 and K stands for Potassium and S stands for Sulphur. The phosphatic fertilizer can be manufactured in various grades like 20:20:00:13, 18:46:00, 14:35 14 etc. Depending on the soil requirements, the usage of these grades will change.

The above fertilizer manufacturing requires ammonia and phosphoric acid for its production.

The various types of manufacturing processes of Phosphoric acid use reaction between a strong acid usually Sulphuric acid and Tri calcium phosphate in rock to give Phosphoric acid and gypsum. The rock phosphate is imported from various sources such as Togo, Jordan, Vietnam, Egypt, Peru, and Morocco. Rock from various sources possesses different chemical constituents and hence the behaviour of these are different. The plant at the IFFCO Paradeep unit has been designed for a blend of 50% Togo and 50% Morocco rocks.

1.2 Plant Production Capacity & Design: -

The capacity of the IFFCO Phosphoric acid plant: is 2650 MTPD (100% PO_4 ,) as a weak acid.

The capacity of Evaporator product: 2250 MTPD (as strong acid)

Annual stream days: 330

The design of the plant was provided by M/s Jacob's Engineering Group, the U.S.A, and its subsidiary, M/s Humphrey & Glasgow Consultants Ltd., provided the detailed Engineering and Construction consulting.





1.3 Product Specifications: -

Components	Strong Acid (% wt.)	Weak Acid (% wt.)
P ₂ O	52.94	27.72
H ₂ SO	3.21	2.02
F	0.98	1.78
CaO	0.05	0.28
MgO	0.79	0.42
Fe ₂ O ₃	1.27	0.67
Al ₂ O ₃	1.14	0.59
SiO ₂	0.20	0.84
Na ₂ O	0.02	0.05
K ₂ O	0.02	0.05
Cl	0.03	0.03
Solid	2.00	1.00

1.4 Physical and Chemical properties of Phosphoric acid: -

Chemical Name: Phosphoric acid.

Classification: Inorganic acid.

Chemical Formula: HPO

Molecular weight: 98

Melting point: 424°C.

Boiling point: 158°C.

Specific gravity: 1.89 at 25°C.

Viscosity: 2.4-9.4 CP (85% aq. Sol.) & 147 CP (100%)

Solubility: Soluble in water.

Chemical stability: Stable.

1.5 Uses of Phosphoric acid:

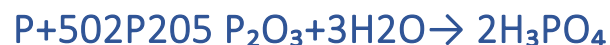
1. Mainly produced for use as a major input in DAP/NPK plants for the manufacturing of Phosphatic fertilizers.
2. Used in the manufacturing of high-grade chemicals, beverages, pharmaceuticals, food products, detergents, and non-fertilizer products.

1.6 Process Technology: -

There are two types of manufacturing processes: -

A. Thermal Process:-

- The raw material used for the production of phosphoric acid by the thermal process is elemental (yellow) phosphorus, air, and water. The thermal process of manufacturing involves three major steps - Combustion, Hydration, and Demisting.
- In combustion, the liquid elemental phosphorus is burned (oxidized) in ambient air in a combustion chamber at temperatures of 1650°C to 2760°C to form phosphorus pentoxide. This phosphorus pent oxide is then hydrated with dilute H₃PO₄, or water to produce a strong phosphoric acid liquid. The final step demisting removes the phosphoric acid mist from the combustion gas stream before released into the atmosphere. This is usually done with high- pressure drop demisters.



- The concentration of H₃PO₄, produced from the thermal process normally ranges from 75 to 85%. This high concentration is required for high-grade chemical production and other non- fertilizer product manufacturing. The efficient plant recovers about 99.9% of the elemental phosphorus burned as phosphoric acid.
- The major source of emissions from the thermal process is H₃PO₄ mist contained in the gas stream from the hydrator. The particle size of the acid mist ranges from 1.4 to 26 micrometres. It is not uncommon for as much as half of the total

PUOs to be present as liquid phosphoric acid particles suspended in the gas stream. Efficient plants are economically motivated to control this potential loss. Although this process can be used with lower-grade phosphate rock, this process has been abandoned because of high electric energy prices.

B. Wet Process:-

The wet process accounts for over 90% of the current Phosphoric Acid production. In the wet process, phosphoric acid is produced by reacting sulphuric acid (H_2SO_4) with naturally occurring phosphate rock. The phosphate rock is dried, crushed, and then continuously fed into the reactor along with sulphuric acid. Rock is dried, crushed, and then from the phosphate, considering sulphate, forming calcium sulphate (CaSO_4), commonly referred to as gypsum. Considerable heat is generated in the reactor. In older plants, this heat is removed by blowing

air over the hot slurry surface. Modern plants vacuum flash cool a portion of the slurry and then recycle it back into the reactor. Gypsum is separated from the reaction solution by filtration. During the reaction, gypsum crystals are precipitated and separated from the acid by filtration. The separate crystals must be washed thoroughly to yield at least a 99 percent recovery of the filtered phosphoric acid. After washing, the gypsum slurry is pumped into a gypsum pond for storage. Water is siphoned off and recycled through a surge cooling pond to the phosphoric acid process.

Wet process phosphoric acid normally contains 26 to 30% POOs. In most cases, the acid must be further concentrated to meet the phosphate feed material specifications for fertilizer production. Depending on the type of fertilizer to be produced, phosphoric acid is usually concentrated to 49 to 52% POs using vacuum evaporators.

In the wet process, there are several processes like: -

- a) Di-hydrate Process (Gypsum as $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$)
- b) Hemi-hydrate Process (Gypsum as $\text{CaSO}_4 \cdot 0.5\text{H}_2\text{O}$)
- c) Hemi-hydrate-Di-hydrate Process
- d) Di-hydrate Hemi-hydrate Process

e) Anhydrite process (Gypsum as CaSO₄)

Modifications are carried out on Di-hydrate / Hemi- hydrate processes and so Hemi-Di hydrate and Di Hemi-hydrate processes were developed which give very good recovery (98- 99%) but their capital cost is high because of the installation of additional Reaction and Filtration system. The production of wet phosphoric acid generates a considerable quantity of acidic cooling water with high concentrations of Phosphorus and fluoride. The excess water is collected in cooling ponds that are used to temporarily store excess precipitation for subsequent evaporation and to allow recirculation of the water to the plant for re-use. Excess rainfall also results in water overflows from settling ponds. However, cooling water can be treated to an acceptable level of Phosphorus and fluoride if the discharge is necessary.

C. Di-hydrate Process: -

This was the conventional process for most of the 20th century. This process has a long track record of reliable operation, but it lacks energy efficiency and many of the operating advantages of the Hemi hydrate process.

Phosphate rocks must be finely ground before processing. The filter product Phosphoric Acid is typically only 25-28% PEOs, so further concentration is required before making phosphate fertilizers.

In this process, rock slurry at 65% solid is metered in the first zone of the annular section where it is diluted with the streams re-circulated from the Cooler Feed Tank. The dilution effect is large because the rock flow is much smaller than the recirculating stream. The rock must be dispersed well to prevent rock coating and to minimize phosphate substitution losses.

Here 1st reaction takes place and a solution of Mono Calcium Phosphate is formed.



Sulphuric Acid is metered through mixing tee with dilute phosphoric acid, coming from the filtration section, to facilitate its dispersion. High local concentrations of Sulphuric acid are undesirable because they cause nucleation and crystal dehydration.

The 2th reaction takes place between sulphuric acid and mono calcium phosphate to form Phosphoric Acid and Gypsum.



Mono Calcium Phosphate + Sulfuric Acid + Water Gypsum + Phosphoric Acid

The di-hydrate process produces gypsum in the form of calcium sulfate with 2 molecules of water (H_2O) i.e., $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (Calcium sulphate dihydrate).

Advantages of Di-hydrate Process:

- (1) A long track record of experience
- (2) Predictable performance

The various technologies of the Di-hydrate process are mentioned below.

- 1. M/s Jacob Process
- 2. M/s Dorr Oliver Process
- 3. M/s Prayon Process

D. Hemi-hydrate Process: -

Hemi-hydrate Process becomes the preferred process for making Phosphoric Acid in the 21st century.

Advantages of Hemi-hydrate Process:

- (1) Minimum capital cost.
- (2) Energy benefits from needing little or no steam to concentrated acid.
- (3) Higher POOs concentration.
- (4) Eliminate 26-42% evaporators.
- (5) Usually eliminate rock grinding.
- (6) Low cooling water requirement.

Hemi-hydrate process products calcium sulphate with a half molecule of water ($\text{CaSO}_2\text{H}_2\text{O}$)

The various technologies of the Hemi-hydrate process are mentioned below.

1. M/s Nissan process.

1.7 Sources of rock phosphate and Effect of their Chemical constituents: -

Rock phosphate is the major raw material used widely for the manufacturing of Phosphoric acid. Rock phosphate contains many chemical constituents. Rock from various sources behaves differently and hence it is usually termed as a living ore. Bone Phosphate of lime (BPL) is the trade name used widely to distinguish rock phosphates based on the % weight of PROs content. The rock phosphate as mentioned above is imported from various sources in the world.

Characteristics of Rock Phosphate:

- A) Phosphates occur mostly in the earth's crust as the calcium component of Orthophosphoric acid.
- B) It ranks as the 11th element out of 115 elements present known.

C) Occurrence of phosphates deposits Phosphates ores are present as sedimentary & igneous.

D) Sedimentary rocks present 85% of the total rock production. E) Igneous rock is present as 15% of the total rock production.

Relation between BPL and PO

BPL (Bone Phosphate of lime) P_2O_5 , 2.183

Orders of Rock Phosphate:

The three grades of rock phosphate are mentioned below in the following table-

<u>Serial Number</u>	<u>Grade of rock Phosphate</u>	<u>% Range of P₂O₅</u>	<u>BPL value</u>
1.	High-Grade Rock	33-39%	73-85
2.	Medium Grade Rock	30-33%	65-72
3.	Low-Grade Rock	26-30%	57-65

Effect of chemical constituents on rock phosphate:

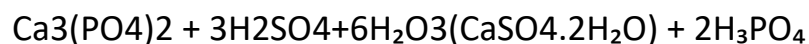
<u>Constituents</u>	<u>Effects</u>
CaO	Affects the sulphuric acid consumption. Each percent of CaO needs an equivalent of 17.5 kg H_2SO_4 per ton of rock. High CaO has an economic effect only. Relatively pure acid can be produced from ore containing very large amounts of calcite.
Fluorine	Occurs as 10% of P_2O_5 by weight in sedimentary rocks. Can be corrosive if not enough reactive silica is also present
SiO ₂	Like quartz, it is only dead weight. Reactive silica (from clays) is needed to combine with fluorides to prevent corrosion. On other hand, higher reactive silica will depress filtration rates.

Al ₂ O ₃ and Fe ₂ O ₃	$0.02 < (A_{\text{Al}_2\text{O}_3} + F_{\text{Fe}_2\text{O}_3}) / P_{\text{O}_3} < 0.08$ Lower alumina content increases corrosion. Increased iron oxide content poses an aftereffect in DAP/NP plant forming water-insoluble components in phosphate fertilizer due to sludge formation in concentrated acid
MgO	Stays with acid phase. Increases viscosity strongly. $\text{Mg}(\text{NH}_4)_2\text{P}_2\text{O}_7$ formation when the acid is ammoniated
NaO	Precipitates as Na_2SiF_6 from the acid. Silicates cause hard scaling on heat exchanger tubes impairing heat transfer.
K ₂ O	Precipitates as KOSiF ₆ from the acid. Silicates are hard to be dissolved with even Sulphuric acid boil out. Impair heat transfer.
SO ₃	Only a dead matter. Slightly saves the corresponding Sulphuric acid consumption
Organics	Causes foaming during the reaction
CO ₂	Higher CO ₂ content causes foaming and unnecessary carryover with the flash cooler operation
Trace elements	Effects crystallization. Have toxic effects in specific final products (Animal feed or fertilizer).

1.8 Process Chemistry:

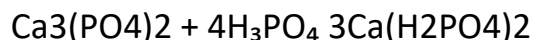
The main reaction in the phosphoric acid plant is the reaction between the Tri calcium phosphate in the phosphate rock and sulfuric acid to give phosphoric acid and gypsum according to equation 1. which occurs in two steps, Phosphoric acid first dissolves the rock in the digestion reaction to form a solution of mono calcium phosphate, Equation 1a. The mono calcium phosphate then reacts with sulfuric acid to form phosphoric acid and gypsum, as depicted in Equation 1b.

Equation: 1



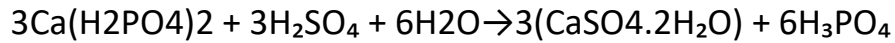
Tri-calcium Phosphate + Sulfuric Acid + Water → Gypsum + Phosphoric acid

Equation: 1a



Tricalcium Phosphate + Phosphoric Acid Mono-calcium Phosphate

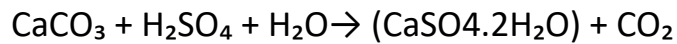
Equation: 1b



Mono-calcium Phosphate + Sulfuric Acid + Water Gypsum + Phosphoric acid

Calcium carbonate in the rock reacts with sulfuric acid to form gypsum and carbon dioxide, equation (2). Most of the carbon dioxide is evolved in the annulus of the Single Tank Reactor.

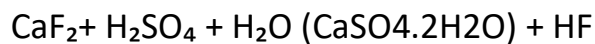
Equation:2



Calcium carbonate + Sulphuric acid + Water Gypsum + Carbon dioxide

Calcium fluoride in the rock reacts with sulphuric acid to form Hydrofluoric acid as in equation 3 which in turn reacts almost immediately with the silica present to form fluosilicic acid in equation 4.

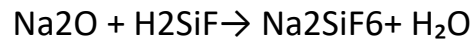
Equation: 3



Calcium fluoride + Sulphuric acid + Water → Gypsum+ Hydrofluoric acid

Equation: 4

Fluosilicic acid reacts with oxides such as sodium oxide, potassium oxide, or magnesium oxide to form fluosilicates. The reaction is illustrated with sodium oxide in equation 5.

Equation :5

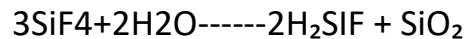
Sodium oxide + Fluosilicic acid → Sodium Fluosilicate + Water

Because the solubility of sodium and potassium fluosilicate decreases rapidly as temperature is lowered, these compounds are prone to form scale in the piping and the equipment downstream of the Reactor.

Some fluosilicic acid decomposes into silicon tetrafluoride in the Reactor and in the Vacuum Cooler, equation 6, and is re-converted by water to fluosilicic acid and silica in the Fume Scrubber and the Vacuum Cooler Pre-condenser, equation 7.

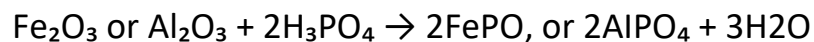
Equation:6

Fluosilicic acid + Heat & Acid → Silicon Tetrafluoride (gas) + Hydro fluoric Acid

Equation: 7

Silicon Tetra fluoride + Water → Fluosilicic acid + Silica

Iron and Aluminium oxide is converted to the corresponding phosphates as in equation 8.

Equation:8

Iron or Aluminium oxide+ Phosphoric acid → Ferric or Aluminium Phosphate -Water

The process of rock digestion is represented in Equation-1a. The reaction rate increases directly with temperature, the content of the particles. Since this area is inversely proportional to particle size the rate increases as particle

size decreases. The effect is however balanced by finer gypsum crystals caused by higher nucleation because more dissolution increases supersaturation.

The reaction slurry must be supersaturated with gypsum for gypsum to crystallize, as is discussed later. Supersaturation can result in the precipitation of calcium sulphate on the surface of the rock particles. This precipitation increases with supersaturation and consequently with sulphate level. Partial coating only retards the dissolution of the rock. The complete coating protects the particle against acid attack and results in P_2O_5 yield loss. This loss is part of the loss measured by the Citrate Insoluble (CI) PDOs test. The digestion rate is very sensitive to sulphate level, particularly at low rock conversion. In a reactor not buffered by a large recirculation flow even minor sulphate increases from acid, flow fluctuations would slowdown dissolution appreciably.

1.9 Site Data: -

Dry bulb temperature: 41°C, 8.9°C min, 35°C average

Wet bulb temperature: 30°C

Relative humidity: 90% max, 50% min, 75% average

Pressure: 1015m bar

Mean annual rainfall: 150cm (average)

Wind velocity: 26 km/hr max.

Prevailing wind direction: SW to NE

Site elevation: 6 m above mean sea level, 4.5m above Admiralty Datum

1.10 Geological conditions: -

Earth quake: The plant site lies within Seismic zone III with an expected seismic intensity of VII on the modified Mercalli scale, 1931.

SECTION-2

PROCESS AND EQUIPMENT

2.1 Detailed Process Description: -

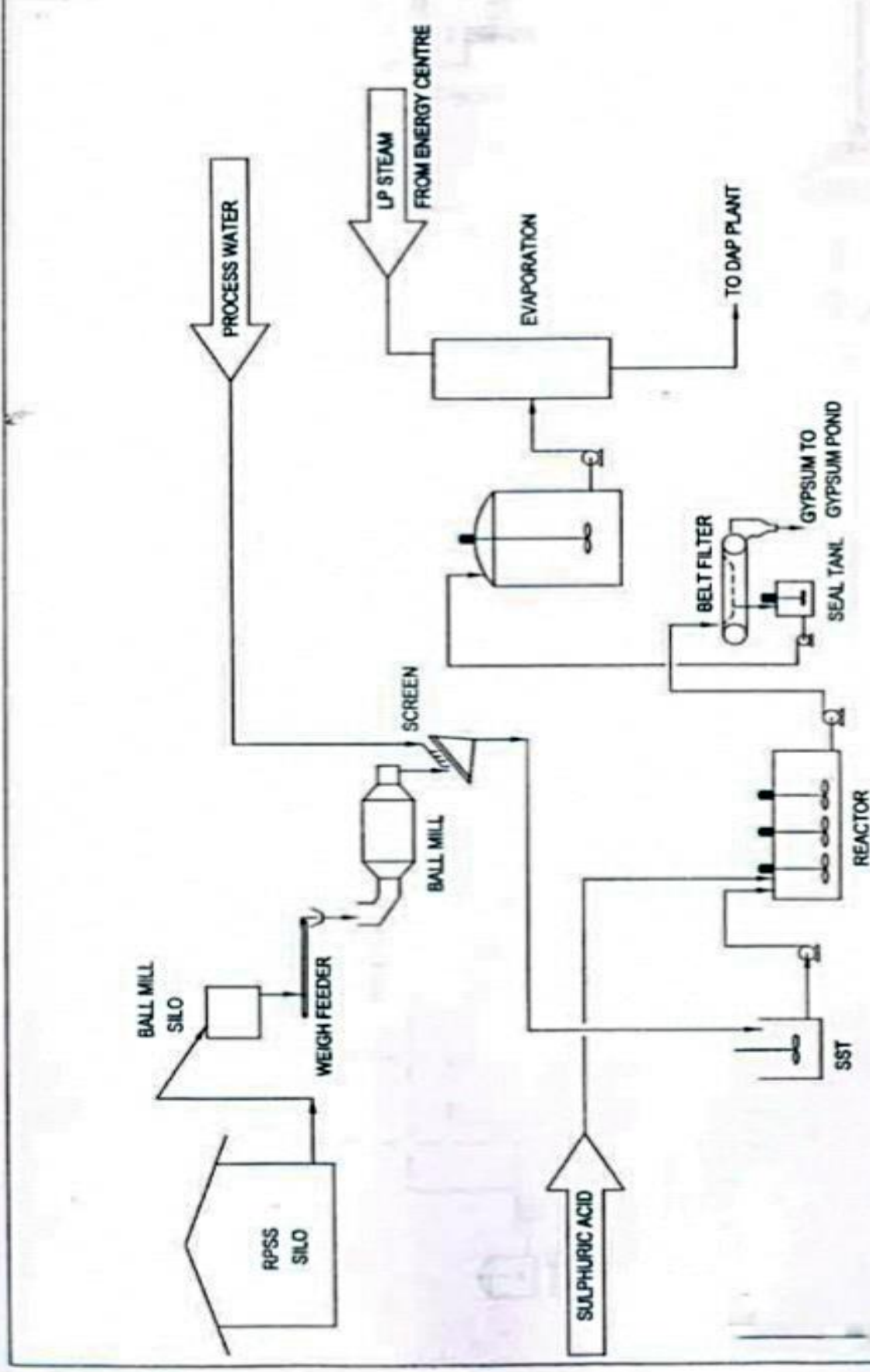
General Features:

Phosphoric acid plant at IFFCO Paradeep unit is designed by M/s Jacob Engineering. The process is Di-hydrate stream equipped with wet grinding and flash cooling. The plant is designed to produce 2650MT of PDO, as weak phosphoric acid (26-27% P_2O_3) of which 2250 MT will be concentrated to strong acid (48-49% POOs) and 161 MTPD of 100% H:SIF (Fluorosilicic acid) is made by recovering the Fluorine released during the evaporation stage using the conventional absorption process. The plant produces 13,250 MT of dry gypsum at rated capacity. Gypsum is pumped to battery limits as 25% (wt.) solids slurry.

The process description divided into the following main sections-

- A. Rock Phosphate storage and conveying section
- B. Grinding Section
- C. Reaction and Fume scrubber section
- D. Belt filter and Filtration section
- E. Concentration and Fluorine recovery
- F. Acid storage and Clarifier section
- G. Cooling Tower
- H. Export Tank
- I. Gypsum Pond

PROCESS FLOW DIAGRAM OF PHOSPHORIC ACID PLANT



PROCESS FLOW DIAGRAM OF PHOSPHORIC ACID PLANT

A. Rock Phosphate storage and Conveying section:

Three rock phosphate storage sheds A, B and C capacities for storing rock phosphate exists. The storage sheds A, B have 75,000 MT each and C has 100,000 MT capacity. Each storage shed built with an underground (tunnel) conveyors 103 A, B & C respectively for conveying rock phosphate from storage to underground silos A, B and present ball mill area. The capacity of each conveyor is 550 MTPH. These conveyors discharge the rock phosphate on conveyor 104 which in turn discharges the material on conveyor 105. The conveyor 105 discharge can be taken to Silo A and on conveyor 106A. Material from 106A was received by Silo A through 106B conveyor. Finally, material from 106B discharged into Silo C.

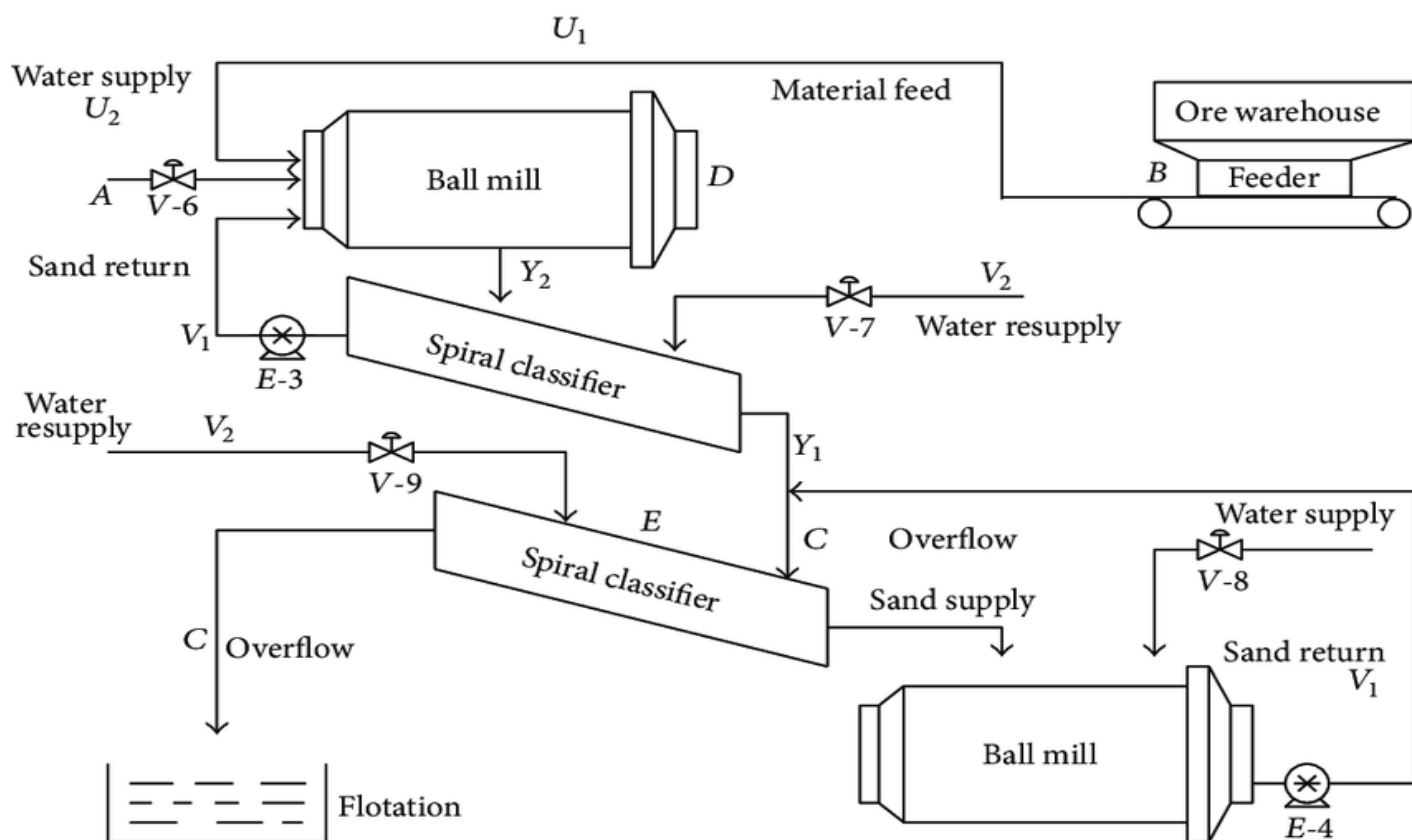
B. Grinding section:

The rock phosphate received in the storage sheds coarser and needs to be crushed to required size in ball mills. There are four ball mills each 110 MTPH capacity. The medium of grinding wet Rock phosphate from battery limits conveyed to two 550 MT and one 900 MT underground rock silos. Silo A supplies rock to ball mill and Silo B supplies feed ball mill and silo C supplies rock to mill. Grinding media balls are high chromium steel balls of 50 mm size.

Each mill operates closed circuit with four Derrick screens. The discharge from ball mill stored in Slurry storage tank providing buffer capacity for ground rock storage. Rock transferred from ball mills from the corresponding weigh feeder along with recycled material from the corresponding rock screens. Water usually treated effluent water from Utilities is fed to ball mill both through the Trommel screen at the discharge end and via the oversize chutes from screens.

A provision for admitting process water as feed to ball mill also exists. Ground rock slurry leaving the ball mill is collected in the corresponding mill slurry tank and pumped to the screens. Oversize material is transferred back to the mill. On specification material (less than 3% of 35 Tyler mesh) is collected as a slurry in the Slurry sludge tank. There is a provision of bypassing the screens and allowing the discharge of mill slurry pumps directly into Slurry sludge tank. The screens are usually bypassed during the processing of Jordan and Egypt rock source. Reactor feed pumps transfer the normal feed to the reactor. There are three reactor feed

pumps of which one in service and two are stand by A reclaim sump with agitator and pump is provided to collect spillage from the milling area for returning spillage to the Slurry sludge tank.



(Fig-Flowchart of the Grinding Section)

C. Reaction and Fume scrubber section:

Rock phosphate slurry and sulphuric acid react in this section to form Phosphoric acid and calcium sulphate di hydrate, $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$ (Gypsum). The reaction is designed to produce a gypsum that can be washed and filtered easily resulting in an expected recovery in the reaction and filtration sections of 96.09(wt.) of the P_2O_5 fed to the plant. Major components in the section are:

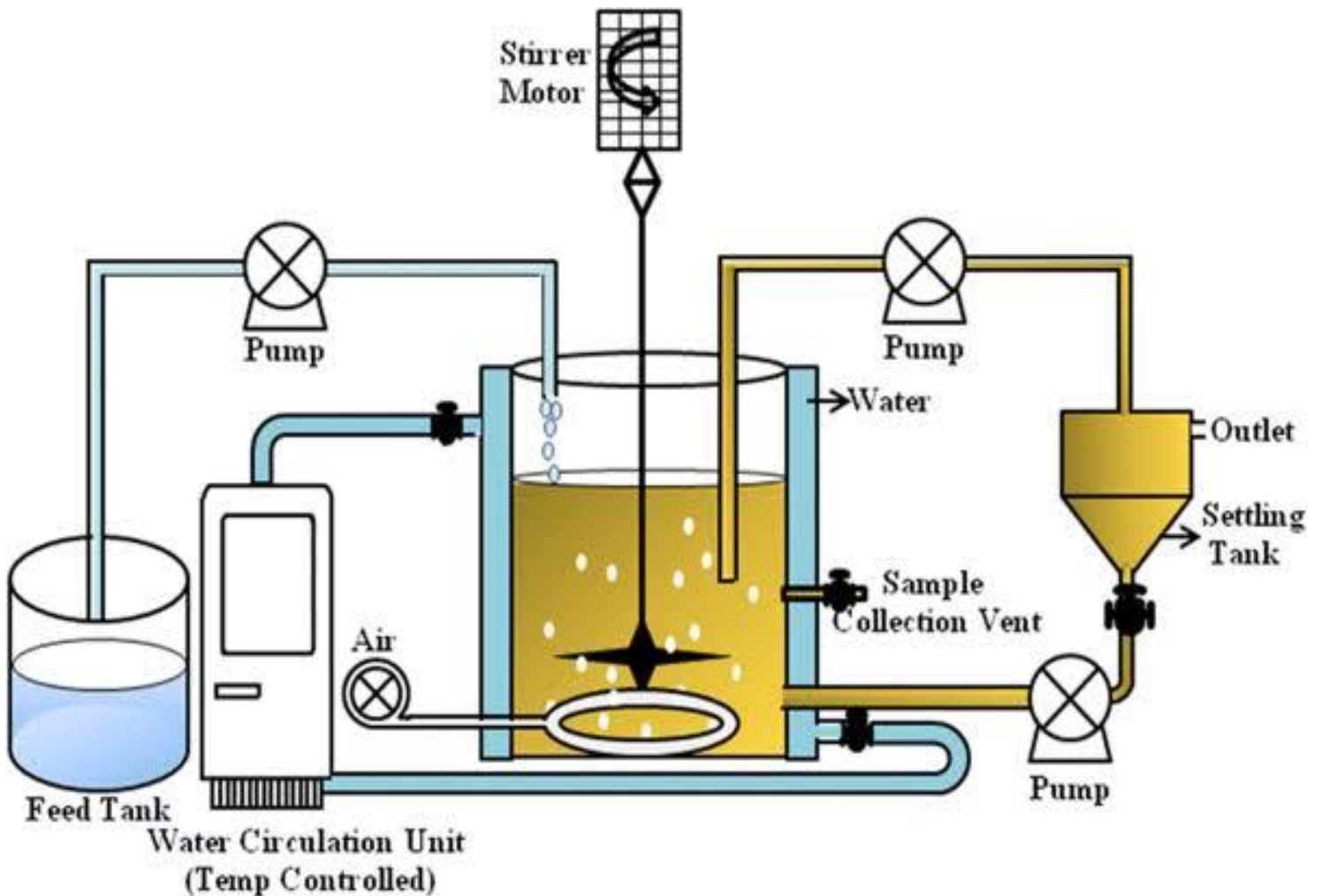
1. Single tank reactor (2300 m³ volume)

2. The adjustment and Filter feed tank (200 m³ volume each)
3. The cooler seal tank (1650 m³ volume)
4. The cooler feed tank (770 m³ volume)

Rock digestion primarily occurs in the single tank reactor with very high slurry. recirculation and mixing. The other vessels provide additional residence time, feed the vacuum cooler, seal the barometric leg of the vacuum cooler and segregate the slurry pumped to the filtration area. Total reaction volume is 5120m³, 2300m³ in the single tank reactor, 770m³ in each of the cooler feed and 1650 m³ cooler seal tank, 200m³ each in the adjustment and filter feed tank. Total residence time with this volume at rated capacity is 3.54 hrs. Four vacuum coolers are equipped with steam ejector system and barometric condenser for creating vacuum. This equipment removes excess heat to control the reaction temperature. Vacuum coolers are equipped with a vacuum cooler circulating pump. These pumps re circulate a total of 36000 m³/hr of slurry from the last vacuum cooler feed section to the first section of the single tank reactor. This large recirculation flow maintains a stable sulphate level by acting as a flywheel against fluctuations, Fume scrubber along with vertical scrubber removes fluorides from plant gas streams before venting to the atmosphere.

- **Reactor:** -

The low level of super saturation results in gypsum crystals that can be filtered and washed easily. The fumes released during the reaction stage are vented from section 117 agitator area to the scrubber. The release of fumes from reactor may form stable foam which can cause problems with the recirculation pumps. Two provisions have been made to control the foam. The top impeller of the agitators in section 111-117 agitator zone is designed to break down the foam mechanically. Addition of defoamer in all compartments of reactor except vacuum seal tank and adjustment tank controls foam in the reactor section. Defoamer is added in reactor to control the foam level as excess foam in reactor will lead to carry over in vacuum cooler circulator. Addition of defoamer is also done in discharge Y piece of vacuum cooler circulator.



(DIAGRAM OF THE CSTR)

1. Single tank reactor:-

The reactor is built as two concrete tanks. The inner tank forms the central compartment. A compartment in the annular area with one eighth of the total annular volume forms the cooler feed tank compartment. The balance of the annulus is taken up by section 111 to 117 agitator zone which is not separated by partitions. Each section is provided with an agitator. Underflow weir connect section 117agitator zone to the centre compartment, the centre compartment to the cooler feed compartment and the cooler feed compartment to the cooler feed tank. Slurry level in the annulus is about 100 mm higher than in the centre compartment, the level in the centre compartment is about 100 mm higher than the level in the cooler feed compartment and the level in the cooler feed

compartment is 100 mm higher than the cooler feed tank. The differences in level, the circulating pumps and the weirs routes the flow from annulus 111 to 117 agitator zone in the annulus, to the centre compartment, to the cooler feed compartment and to the cooler feed tank. Most of the flow 34800 out of total of 36000 m³/hr is re-circulated from the cooler feed tank back to section 11 agitator. Rock slurry at 65% solids is metered into section 1 where it is diluted with the streams re circulated from the cooler feed tank. The dilution effect is large because the rock flow is much smaller than the re circulating stream at rated capacity. The rock must be dispersed well to prevent rock coating, passivation and substitution losses.

Sulphuric acid, 98% H₂SO₄. is metered and admitted into reactor near agitator 111, 115, 116 via four mixing Tees. The acid is diluted in the mixing tee with return acid recycled from filtration area to facilitate its dispersion. High local concentrations of sulphuric acid are undesirable because they cause nucleation and crystal dehydration. Sulphate level is almost constant throughout the annulus because of large back mixing flow produced by the agitators.

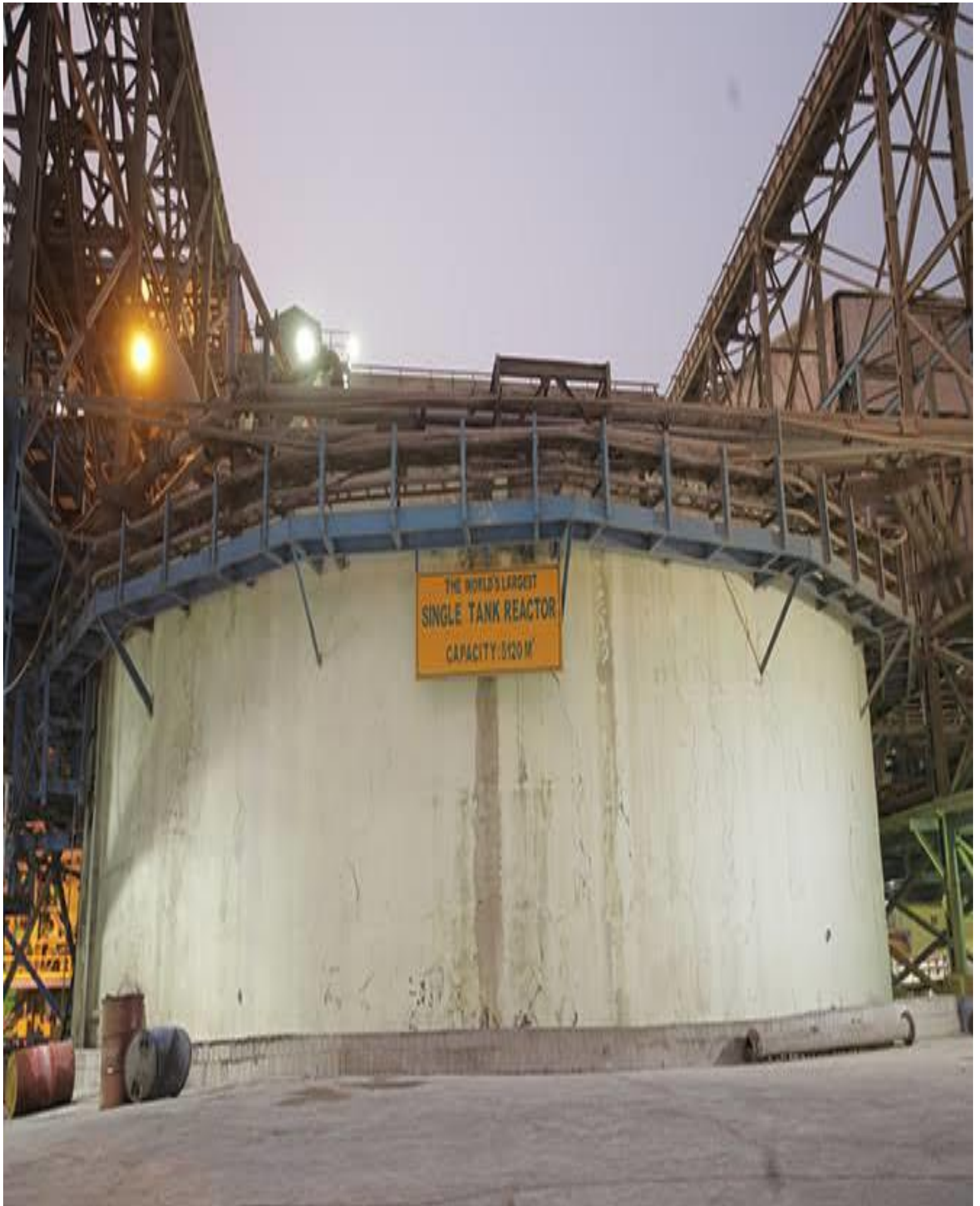
The local sulphate gradients caused by the additions of rock and acid are therefore minimized. The major part of rock digestion and calcium sulphate crystallization occur in the annular sections. Reaction temperature is maintained at 79.5°C -81.8°C to promote formation of di-hydrate crystals and rapid rock digestion. The sulphate level is controlled to obtain low level of super saturation and to minimize insoluble phosphate losses in the gypsum

2. Adjustment and Filter feed tank:

This tank has two compartments-

- 1) the Adjustment section, and
- 2) the Filter feed section.

The adjustment compartment is equipped with a line to add sulphuric acid on flow control to lower the calcium concentration in the filter feed liquor without affecting the sulphate level in the single tank reactor.



3. Cooler seal tank:-

The cooler seal tank is a rectangular concrete tank identical in size to the cooler feed tank. The tank seals the down leg of the vacuum coolers. It overflows through a weir to the annular section of the single tank reactor and to the adjustment compartment of the Adjustment and filter feed tank. Since weir to the reactor is lower than the weir to the adjustment compartment, the slurry flows preferentially to the reactor for recirculation. This tank is vented to the fume scrubber.

Vacuum cooler:

The reaction in the single tank reactor is exothermic. The vacuum coolers remove the heat to control reaction temperature by evaporating a small amount of water from the re circulation slurry under a vacuum of about 11.9" Hg. Cooling also de supersaturates the filter feed to reduce scaling of the filter. The coolers are mounted directly on the cooler feed tank to minimize the head of the cooler circulating pumps. A head of slurry above the inlet nozzle prevents flashing and scaling in the inlet line. Normally 100% of the total recirculation flow, 36000m³/hr, is pumped through the coolers with the vacuum cooler circulating, 9000 m³/hr each. The cooler is designed to limit slurry cooling to about 2.7°C to minimize nucleation and scaling. Each vacuum cooler has two barometric condensers, the vacuum cooler pre condenser and the vacuum cooler condenser and a vacuum cooler ejector. The ejector uses medium pressure steam as the motive fluid.

The Pre condenser removes more than 90% of the fluorine released in the vacuum cooler by absorption in cooling tower. The barometric leg of the pre condensers are sealed in vacuum pre condenser seal tank which is vented to the fume scrubber. The V.C pre condenser pumps re-circulate the water from the tank through the spray nozzles in the pre condensers. The system recovers part of the heat of reaction by heating the cooling water from 35°C to 65°C.

The vacuum pre condenser seal tank supplies water to the filtration area for cake, cloth and belt washing. To ensure that the water is at the temperature required in the filtration area, about 65°C, low pressure steam is sparged into the seal tank. A temperature controller regulates the steam flow.

The balance of the vacuum vapours about 80% of the total is condensed with reused cooling tower water in the vacuum cooler condensers, cascade type barometric condensers. The barometric leg of the condensers is sealed in a tank, vacuum cooler condenser seal tank, which is vented to the fume scrubber. The tank overflows continuously to the cooling tower.

Non condensable from the Vacuum cooler condensers are removed by the steam ejectors. Since the ejector's discharge is piped to the seal tank, non-condensable are eventually vented to the Fume scrubber. Air is bled into the vapor line downstream from the vacuum cooler condenser to control pressure in the vacuum cooler.

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4. Cooler feed tank:

The cooler feed tank is a rectangular concrete tank that is connected to the cooler feed compartment of the single tank reactor system by an underflow. Vacuum cooler circulating pumps feed slurry from this tank to each vacuum cooler. This tank is vented to the fume's scrubber.

- **Fume scrubber:**

A plant wide header routes fumes to the Fume scrubber. The scrubber reduces the fluoride level in the gas vented through the plant stack to 10g as Fluorine per ton of POs fed to the plant

The fluorine-laden vapor flows horizontally through vertical scrubber where 8 no. nozzles arranged in each of three rows sprays water to absorb fluorine fumes. The packing stages are washed with cooling tower water through large orifice, low pressure drop, spray nozzles. To prevent deposition of silica the vapor is washed with sprays

The scrubber operates at a slight vacuum from about -100 mm H₂O (g) at the inlet to -350mm H₂O (g) at the outlet provided by the scrubber exhaust fan. The fan discharges to the plant stack. Spray water drains to the Fume scrubber sump and is pumped by the Fume scrubber sump pumps to the belt filter sluicing header which helps in disposing the gypsum. Hot water from barometric condenser of evaporator is used for spraying purpose in pre scrubber and fume scrubber horizontal chamber.

D. Belt filter and filtration section

- **Belt Filter /Filtration:**

In the filtration section, seven horizontal belt filters (six in service and one stand by) separate acid and gypsum. The filters operate in parallel and when more than one filter is not operating, the plant can continue operation at reduced rate with the other filters. The gypsum is washed three times on the filter to reduce the water soluble P2O₅. The filter design limits P2O₅ dilution across the filter to 0.5% maximum. Filter product containing 26-27% POOs is pumped to the weak acid

clarification section. Washed gypsum is slurried and pumped to the gypsum disposal site.

The gypsum cake formed on the filter cloth travels in series through four flooded compartments, formed by adjustable dams, as an endless rubber belt moves the filter cloth from the feed end to the discharge end of the filter. The liquid in the compartment passes through the cake and the cloth into the channels formed by the ribs of the rubber belt. These channels route the flow to the openings along the belt's centre line, which drain the liquid into one of the five compartments or sections of the vacuum box underneath the belt.

The cloth and the belt return to the front end of the filter through separate but parallel paths. Both sides of the cloth are washed continuously with water from cloth wash tank. to prevent blinding. The belt is also washed with water from this tank, intermittently, as needed to remove scale-build up.

The warm water from the vacuum cooler pre condenser seal tank is used as filter wash water. The water is received on demand from the spillage sump in filtration area. The wash water flows into the cake wash tank through spillage sump pump. From the cake wash tank it is pumped on to the filter as filter cake wash.

Each vacuum box section drains to its filtrate separator, then via barometric leg into one of the compartments of either the primary filtrate seal tank or the secondary filtrate seal tank. All filtrate separators are connected through a **header** to a vacuum system, which provides the pressure drop for liquid flow through the cake.

The first flooded compartment on the belt holds product acid. The acid collected from the front end of this compartment is diluted with the cloth wash water retained by the cloth. Furthermore it contains some solids because it drains through cake not fully formed. This

acid drains through the cloudy filtrate section of the vacuum box to the return acid compartment of the primary filtrate seal tank for recycle to the reactor. The balance of the acid from the first flooded compartment drains through the No.1

filtrate section of the vacuum box to the product and compartment of the primary seal

tank. This acid is the 26-27% POOs filter product acid which is pumped to the weak acid storage and clarification section.

The other three flooded compartments on the belt are used for counter current cake washes. The fourth flooded compartment normally holds water from the cake wash water pump. The filtrate drains through the No.4 section of the vacuum box to the No. 4 filtrate compartment of the secondary filtrate tank for use as wash in the third flooded wash compartment

The fourth flooded compartment can be washed additionally with water from final wash sump which receives water from recovery sump. The seal water outlet of the vacuum pump provides the input source to the recovery sump and water flows from recovery trench to final wash sump

The third flooded compartment is washed with No. 4 filtrate. This wash drains through the No. 3 filtrate section of the vacuum box to the No. 3 compartment of the secondary filtrate tank, for use as wash in the second flooded compartment.

The second flooded compartment is washed with No. 3 filtrate. This wash drains through the No. 2 filtrate section of the vacuum box to the return acid compartment of the primary filtrate seal tank for recycle to the reactor.

The recycle flow to the reactor consists of three streams, blended in the return acid compartment of the primary filtrate seal tank, product acid, which overflows from the product acid compartment, cloudy filtrate and No. 2 from the filter.

- **Filtration Section: -**

The first and second flooded compartments in the filter and the two filtrate seal tanks are vented to the fume scrubber.

Gas from the filtrate receivers flows through a vacuum header to an acid trap where entrained liquid is separated. The liquid drains to the No.4 filtrate compartment of the secondary filtrate tank. The gas mainly air saturated with water flows to a packed bed condenser, the filter condenser, where most of the water vapor is condensed with process water. The barometric leg of the condenser is sealed in the recovered water sump. The gas leaving the condenser is compressed with the liquid ring Filter vacuum pump which discharges to the atmosphere. Air is bled in downstream of the condenser to control the vacuum level in the header. Process water/TG blow down water is used as seal water for the pump.

The gypsum leaving the filter is sluiced in the gypsum cake hopper with pond water or blow down water from fume scrubber sump. Gypsum and water flow by gravity in to the gypsum slurry tank where additionally pond water is added to form 25% solid slurry. The slurry pumped to battery limits with the gypsum slurry pumps. Three gypsum pumps exist of which two are service and one is stand by.

E. Concentration and Fluorine recovery:

- **Concentration/Evaporator**

In this section, phosphoric acid is concentrated from 27% to 49% P₂O₅ in the evaporator. The concentrated acid is pumped to the acid storage and clarification section.

There are seven evaporators (six running and one stand by). Each evaporator is a single stage, forced circulation unit consisting of a rubber lined flash chamber, axial flow evaporator circulation pump and vertical shell and tube, evaporator heat exchanger with impervious graphite tubes. Acid is re circulated at a high rate (7.315 m³/hr) by the pump through the heat exchanger and into the flash chamber where an absolute pressure of about 2.6 Hg is maintained to keep the acid at its boiling point. Swirl induced by the tangential entry of acid to the flash chamber and to minimize potential cavitation of the pump, vortex breaker is mounted at the bottom of the conical section of the flash chamber. A cone shaped lump screen of SS 904L is installed a head of the heat exchanger to catch large pieces of scale, rubber or other foreign matter that could block flow through the tubes.

The evaporator heat exchanger is a single pass shell and tube unit consisting of 649 tubes vertically mounted in a carbon steel shell with graphite tube sheets. Each tube is 7315 mm long with an outside diameter of 50.8 mm and inside diameter of 38.1 mm. At the rated pump capacity, the average acid velocity through the tubes is 2.75 metres per second with an acid temperature rise of 2.6°C. The liquid level in the flash chamber is set at a sufficient height to suppress boiling in the tubes which would result in scaling of heat exchanger surface

Feed acid enters the evaporator through a nozzle in the chamber at the discharge end of the heat exchanger. This chamber also has a removable full diameter man way providing access for inspection or in place cleaning of the tubes. Product acid overflow from the flash chamber through side discharge piping which maintains a constant liquid level slightly above the tangential entry.

The lower 1500 mm of the heat exchanger shell is constructed of 316 L SS to provide corrosion resistance in the event of acid contamination of the condensate. A small pressure relief valve is provided on the shell to warn that steam pressure is approaching the maximum allowable. Full pressure relief is usually provided by a larger relief valve mounted on the steam distribution piping upstream of the battery limit supplied by others.

Overhead vapours from flash chamber pass to an Entrainment separator before fluorine recovery. These are rubber lined vessels. Acid removed by entrainment flows back to the suction side of the evaporator circulation pump.

The acid outlet of the heat exchanger is located below the liquid level in the flash chamber to suppress boiling in the tubes. Low pressure steam is supplied through a flow control valve to the shell side of the heat exchanger. Condenser from the heat exchanger flows by gravity through a condensate receiver of SS 316L to a condensate pump which returns it to the battery limits. Evaporator can be drained and emptied through a drain pump which pumps the acid to the strong acid distribution box.

Vacuum is established in the evaporators by admitting cooling tower water into the barometric condenser. Barometric legs from the barometric condenser are sealed in a condenser seal tank constructed of rubber lined steel. The water from the seal

tank flows by gravity to the hot well. A proportion of water entering the seal tank is reused in fume scrubber spray section of reactor. Earlier vacuum is established by also admitting MP steam in the three-stage ejector system connected with barometric condenser. After replacement of baffle trays of barometric condenser with a single perforated tray the ejector system is obsolete in usage as vacuum is achieved only through the admission of cooling tower water in the barometric condenser.

The evaporator heat exchanger gradually becomes fouled due to scaling tendency of the acid. As this occurs it becomes necessary to increase the steam pressure and temperature to maintain the same heat transfer rate. At some point the rate will be reduced to such an extent that it becomes necessary to shut down the evaporator for cleaning. This procedure is referred to as an evaporator boil out and is accomplished by re circulating a dilute solution (approximately 5%) hot Sulphuric acid with pond return water to prepare the boil out solution. At the completion of the boil out the cleaning solution is drained.

The solution is transferred by the evaporator drain pump to the wash water storage section of distribution box.

Sometimes the heat exchanger tubes are so fouled, they become plugged. When this occurs, a high-pressure water pump is used to lance and clean the tubes. The pump is a portable cleaning system that can be moved to the appropriate evaporator for cleaning

The concentrated acid at a temperature of 85°C overflows from the flash chamber to a concentrated acid pump which delivers it to one of the two strong acid clarifiers T-413 & T-414. Collection of the spillages in the evaporator area is possible through a sump with a pump.

- **Fluorine recovery section:**

During the flashing phenomenon, vapours from flash chamber reach the fluorine scrubber where spray of process water through 12 set of nozzles arranged at three different elevations absorb the HF fumes from the vapor. Circulation of 850 m³ of water per hour is established from the seal tank to the fluorine scrubber through a circulation pump.

The fluorine scrubber is essentially a large void spray tower where the gases leaving the evaporator are passed counter current to a large re circulating flow of fluo-silicic sprayed from nozzles at three different elevations. Re circulating liquor from the sprays flows by gravity out the bottom through a barometric seal leg to a fluo-silicic acid tank. The tank serves two functions. One is to provide barometric seal for the down leg of fluorine scrubber and other is serving as a pump tank for the fluorine scrubber circulation pumps. A partition separates the tank into two compartments which serve the dual functions. The tank is vented to the fume scrubber.

Periodically, the density will be checked and product will be transferred to intermediate tank whenever the density of acid reached 1.18/1.19 corresponding to 19.0-20.0% strength. Defoamer is also added in acid feed line to evaporator for suppressing foam generation in flash chamber and thereby preventing carryover of acid.

From the intermediate storage the product is pumped to main FSA storage tank. After completion of batch, the control valve on makeup process water line to seal tank will open and tank is filled to the desired level. The circulation of water is continuous and batch wise the product is transferred.

F. Acid Storage Area and Clarification:

There are two clarifiers each for weak and strong acid storage.

SL.NO	Weak acid clarifier	Volume (m³)	Strong acid clarifier	Volume (m³)
1	T - 411	2100	T - 413	4000
2	T - 412	4200	T - 414	4200
3	T - 410	10000	—	—

The tanks are constructed of rubber lined carbon steel with a layer of carbon brick on the bottom. They are equipped with a rubber covered 317 L SS rake, centre feed y bottom sludge discharge cone and clarified acid take off nozzles at three

elevations well. The rake mechanisms are automatically raised if torque exceeds a pre-set value Sludge pumps provide the circulation of acid in the tank.

Overflow from the weak acid clarifier flows to pump tank of 1100 m volume. Weak acid from pump tank will be pumped to evaporators and DAP as per requirement. Similarly New pump tank serve as pump tank for pumping strong acid to DAP.

The filter acid distribution box is divided into three compartments one compartment towards T-411, other towards T-412 and middle one is wash box zone. Similarly, the strong acid feed distribution box is also divided into three compartments one compartment towards T-413, other towards T-414 and middle one is wash box. The feed hoses are diverted to either one of the boxes as per the requirement.

There are seven feed pumps which cater weak acid feed to the evaporators. The acid transfer to DAP is as tabulated below.

Sl.NO	Weak acid transfer lines to DAP	Strong acid transfer lines to DAP
1	W new 1 line through P 496	Strong new line through P 441,443
2	W new 2 line through P 496	Strong old line through P 442
3	W old line through P 497	—

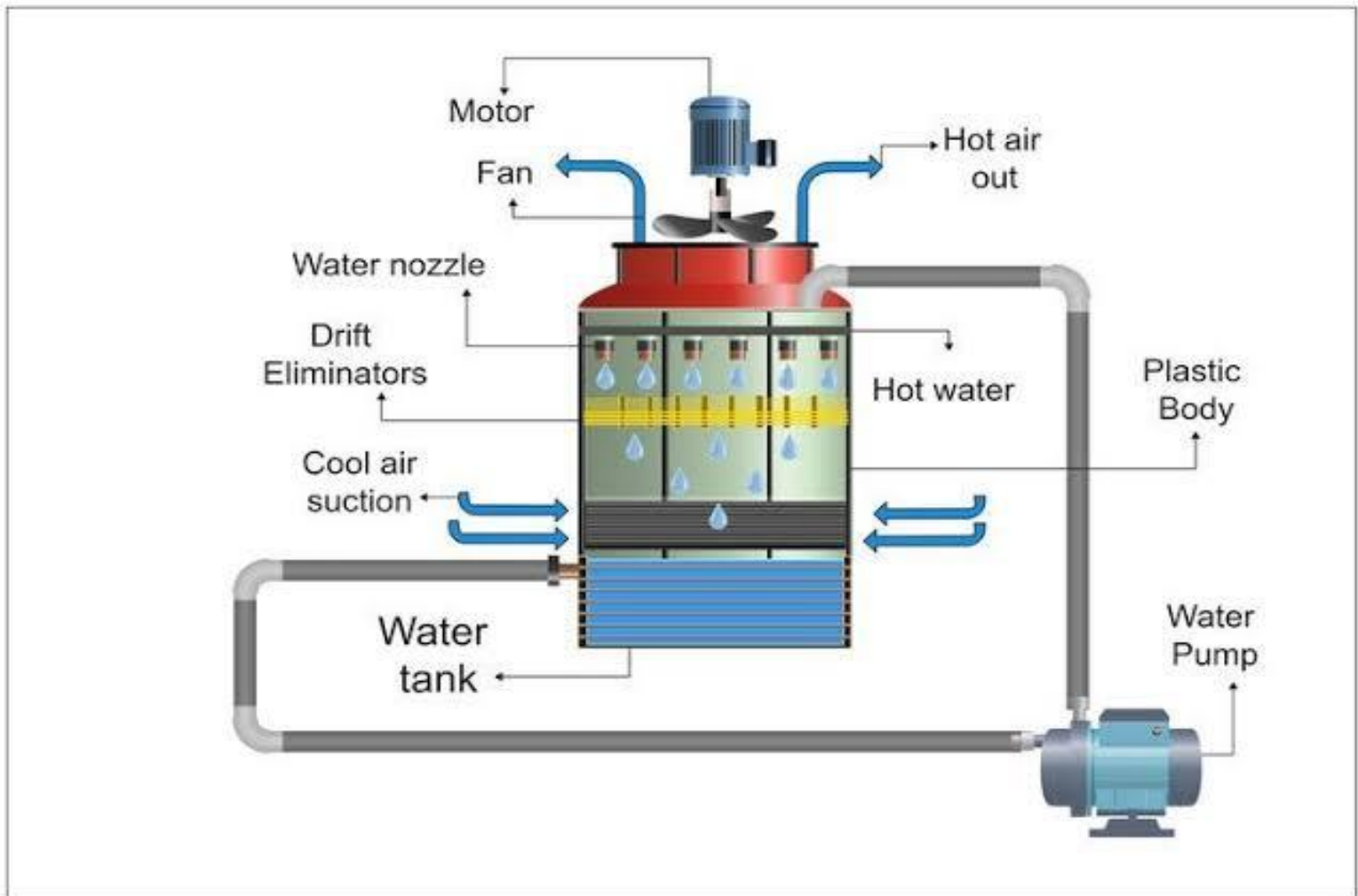
P 442 can cater both weak and strong acid to DAP.

A plate heat exchanger is installed in evaporator section for cooling the strong acid for export to Kandla unit.

G. Cooling Tower:

Hot water from the plant flows by gravity to the cooling tower hot well. There are seven hot well pumps and seven cold well pumps. Each pump is rated for about 4,600 m³/hr. The cooling tower has eight cells each with a stainless steel or glass reinforced vinyl ester fan. The tower has a treated frame with glass reinforced vinyl ester louvers. Filling is PVC on a Glass reinforced Vinyl Ester grid. Cooling water gravitates to the base of the tower which sits above the cold well. There are two

pond water lines for adding make up water to cooling tower. All the fan blades were recently replaced with hollow one as part to conserve energy. The duty of the cooling tower is 27,448 m³/hr @ 30°C WB, In/Out 43/35°C, side entry, and top exit with acid brick line concrete basin



(Flow sheet of the cooling tower)

H. Export Tank:

The export tank is constructed and situated near Ammonia storage area for exporting strong phosphoric acid. The tank is 40m in diameter and 12 m height with three outlet nozzles. Two pumps are provided for transferring the acid to either the

ship or to the strong acid distribution box in PA plant. Pump 444 in PA storage area serves to pump acid from plant to export tank.

Air distributors are provided inside the tank for keeping the solids in suspension. A provision to flush the discharge and suction lines of pumps with air also exists. A sump with a pump to collect the spilled acid back to the export tank also is provided. Pond water line for cleaning and recollection system provision back to PA plant exists.

Gypsum Pond:

There are two ponds for storage of gypsum.

1. East Pond
2. West Pond

Gypsum slurry from the plant is pumped to the pond through gypsum pumps in filtration section. The pond consists of three dykes-

1. The inner dyke
2. The outer dyke, and
3. The rim dyke.

The slurry pumped from the plant flows through the inner dyke towards inside of the stack. Gypsum gets decanted in the middle and water from pond flows out through the rim dyke and enters the outer dyke. The water from the outer dyke enters the surge pond. In the surge pond there are five return water pumps which pump the pond return water to PA plant. As per the plant requirement, pumps will be run. The water level of the surge pond will be monitored on shift basis. In addition to the water from the gypsum stack,

surge pond receives water from garland dyke which collects the rain water in the nearby vicinity.

The selection of pumping slurry to the desired pond will depend on the water level over the gypsum pond and stacking activity. Heavy equipment like excavators, cranes, dozer operate to pile up gypsum stacking.

CONCLUSION

The training period in IFFCO has given me an exposure into the industrial world. The maintenance work showed the difference between theory and practicality. The working of various equipments like Ball Mill, Single Tank Reactor, Belt Filter, Clarifier, Evaporation was learnt and it was a great experience to see these machinery working.

The training will surely help me in my future.